Probability – 2

Geometric probability and random points

Homework

11 сентября 2026 года

Theoretical minimum. Uniform distribution and probability as a ratio of measures; conditioning to reduce dimension; exchangeability and counting cyclic orders; the Crofton measure on lines and Cauchy's formula for the perimeter, and its form in $\mathbb{R}^d$: for a uniformly random unit vector u and any $y \in \mathbb{R}^d$ one has

$$\mathbb{E}|\langle u, y\rangle| = \kappa_d \|y\|, \qquad \kappa_d = \frac{\Gamma(d/2)}{\sqrt{\pi}\,\Gamma((d+1)/2)} > 0$$

(rotation invariance and homogeneity), which reduces statements about $\|y\|$ to one-dimensional ones; linearity of expectation applied to indicators of "a given point is a vertex of the hull".

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1. (Putnam 2021) The plane is tiled by an infinite checkerboard of unit squares, and another unit square is dropped at random, with position and orientation independent of the tiling. What is the probability that it covers no corner of the tiling?

2. (Высшая лига 2023) A point A is chosen uniformly inside the unit cube in $\mathbb{R}^3$. Find the expected distance from A to the surface of the cube.

3. (Putnam 2005) Let $n \geqslant 4$ and let $P_1, \ldots, P_n$ be chosen independently and uniformly on a circle. What is the probability that at least one of the vertex angles of the convex polygon $P_1 \ldots P_n$ is acute?

4. (Сибирская 2024) Points A, B and O are chosen independently on a circle of radius R. Let AB be the shorter of the two arcs determined by

A and B, and let C_1C_2 be the arc of the given circle inside the circle of radius $r = \frac{\pi R}{2}$ centred at O (so $OC_1 = OC_2 = r$). Find the probability that the arcs AB and C_1C_2 do not meet.

5. (Proposed problems) Show that the probability that a point chosen uniformly in $[0,1]^3$ lies at distance at most $\sqrt{2}$ from the origin equals $\frac{(15-8\sqrt{2})\pi}{12}$, and deduce that $\int_0^{\pi/4} \frac{\cos^{3/2} 2\theta}{\cos^3 \theta}\, d\theta = \frac{(4\sqrt{2}-5)\pi}{4}$.

6. (MEXMAT, 2026) In $\mathbb{R}^6$ let A and C be points and B the midpoint of AC. Consider the spheres S_A, S_B, S_C with these centres and radius AB. Choose three points on S_A, one on S_B and two on S_C, independently and uniformly with respect to surface measure. What is the probability that the hyperplane through these six points meets the segment AC?

7. Solve Sylvester's four-point problem: compute the probability that four points chosen uniformly in a triangle form a convex quadrilateral, and do the same for a disc.

8. (IMS 2022) Let Δ be the set of (p_1, p_2, p_3) with non-negative coordinates summing to one, and let $v_1, v_2, v_3 \in \mathbb{R}^3$ with v_1 of maximal Euclidean length. If a point of Δ is chosen uniformly at random, prove that $\Pr\big(\|p_1v_1 + p_2v_2 + p_3v_3\| \leqslant c\|v_1\|\big) \leqslant 4c$ for every $c \in (0, \frac{1}{4})$.

9. Prove that the expected area of the convex hull of n points chosen uniformly in a convex body $K \subset \mathbb{R}^2$ tends to the area of K, and compare the rate for a disc and for a polygon.

10. Prove that for n points chosen uniformly on a circle the probability that they all lie in some semicircle equals $n2^{1-n}$, and find the analogue for a half-space in $\mathbb{R}^d$.